15 A solid cube is floating in equilibrium in liquid mercury. The cube is made of iron of density $7900\,\text{kg}\,\text{m}^{-3}$.

The cube floats with 42% of its volume above the surface of the mercury.

What is the density of the mercury?

A $3300\,\text{kg}\,\text{m}^{-3}$

B $4600\,\text{kg}\,\text{m}^{-3}$

C $14\,000\,\text{kg}\,\text{m}^{-3}$

D $19\,000\,\text{kg}\,\text{m}^{-3}$